FINAL PROJECT Report: Is the Juice Worth the Squeeze? An Educated Guess on the Payoff of Academia

Aldiyar Abzhan, Mia Hower - Group E

December 5, 2025

Abstract: The present investigation explored which institutional characteristics of U.S. colleges and universities best predict graduates' median earnings ten years after enrollment. Given the rising cost of higher education, increasing student loan debt, and ongoing debate surrounding the economic value of a college degree, this analysis seeks to clarify which school-level factors are associated with stronger long-term financial outcomes for graduates. Using College Scorecard data from 2014, we examined seven institutional variables: admissions rate, median verbal and quantitative SAT scores, total undergraduate enrollment, average cost of attendance, student–faculty ratio, and median family income. Evaluating candidate models using Mallows' Cp, parsimony, and considerations of multicollinearity, we determined that a three-predictor model including average cost of attendance, median SAT Math score, and median family income best predicted median earnings ten years after enrollment. With an adjusted R^2 of 0.513, this model accounts for approximately 51.3% of the variability in graduates' median earnings.

Background & Meaning:

Over the past two decades, the cost of attending college in the United States has risen sharply, outpacing both inflation and median household income (Coughter & Gowder, 2024). As a result, students and families increasingly rely on federal financial aid and student loans to finance higher education. This has contributed to historically high levels of student debt, which are approaching 1.8 trillion nationally (Hanson, 2025). At the same time, the economic return on a college degree has become more heterogeneous across institutions, and graduates from some colleges experience substantially stronger earnings outcomes than others. This divergence has intensified debates about whether all postsecondary institutions provide sufficient economic value to justify their costs.

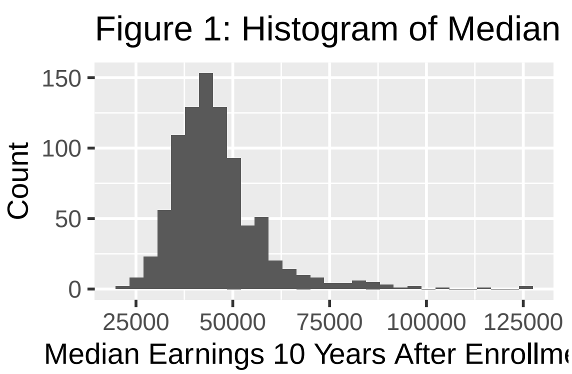
In response to these concerns, federal policymakers have begun to place greater emphasis on transparency and accountability in higher education financing. In recent weeks, the U.S. Department of Education implemented new FAFSA warning indicators for institutions and programs whose graduates earn less than someone with a high school diploma only. These warnings are designed to inform prospective students about the potential financial risks associated with enrolling in certain colleges or programs (U.S. Department of Education, 2025). While these policy changes reflect a growing reliance on institutional-level earnings outcomes as indicators of educational value, there remains limited empirical clarity regarding which specific institutional characteristics most strongly predict long-term earnings.

Prior research suggests that factors such as academic selectivity, student socioeconomic background, and institutional resources are associated with post-graduation earnings, yet these characteristics are often highly correlated, which complicate the efforts to disentangle their independent effects (U.S. Department of Education, 2025). Clarifying how these factors jointly relate to earnings is particularly important given the growing reliance on earnings-based metrics in financial aid policy. By isolating a parsimonious set of predictors, this analysis provides empirical context for FAFSA warning policies and contributes to broader discussions about how the economic value of higher education is assessed and communicated to students.

Based on prior literature, we hypothesized that: Median math and verbal SAT scores, median family incomes, and the cost of tuition are positively correlated with median earnings ten years after entering the institution.

School acceptance rates, and student to faculty ratio are negatively associated with median earnings ten years after entering the institution.

```r
library(ggplot2)

set.seed(230)

n <- nrow(plot_data_complete)

train_index <- sample(1:n, 0.75 * n)
test_index <- setdiff(1:n, train_index)

college_train <- slice(plot_data_complete, train_index)
college_test  <- slice(plot_data_complete, test_index)

# Figure 1: Histogram of Median Earnings
ggplot(college_train, aes(x = MD_EARN_WNE_P10)) +
  geom_histogram(bins = 30) +
  labs(title = "Figure 1: Histogram of Median Earnings (Training Set)",
       x = "Median Earnings 10 Years After Enrollment",
       y = "Count")
```




```r
library(tidyverse)

vars <- c("MD_EARN_WNE_P10", "MD_FAMINC", "STUFACR", "ADM_RATE_SUPP",
          "SATVRMID", "SATMTMID", "COSTT4_A", "UGDS")

plot_long <- college_train %>%
  select(all_of(vars)) %>%
  pivot_longer(cols = everything(), names_to = "Variable", values_to = "Value")

# Table 1: Summary Statistics for Median Earnings
favstats(~ MD_EARN_WNE_P10, data = college_train)
```

```
##    min    Q1 median    Q3    max     mean       sd   n missing
##  20600 38400  43600 50100 124700 45821.73 12076.79 879       0
```

Method:

Data: Data for the present analysis were drawn from the U.S. Department of Education's College Scorecard dataset (2014-15). The College Scorecard provides institution-level information on admissions characteristics, student demographics, costs, and post-graduation earnings for U.S. postsecondary institutions that participate in federal financial aid programs. Although the full dataset includes nearly 6,000 institutions, many schools did not report complete information for our key variables of interest. As a result, the analytic sample was restricted to institutions with complete data on all selected predictors and the response variable. The final analytic sample consisted of 1,173 institutions, which was randomly split into a training set (N = 879) and a test set (N = 294) for model evaluation and prediction. Because missingness in the College Scorecard data is unlikely to be random, and schools with limited reporting capacity or open admissions policies may be underrepresented, results should be interpreted as applying primarily to institutions that report comprehensive admissions and outcome metrics. Additionally, after 2014, the College Scorecard dataset stopped reporting graduate median earnings 10 years after enrollment, so our analysis may be out of date or adjusted for inflation.

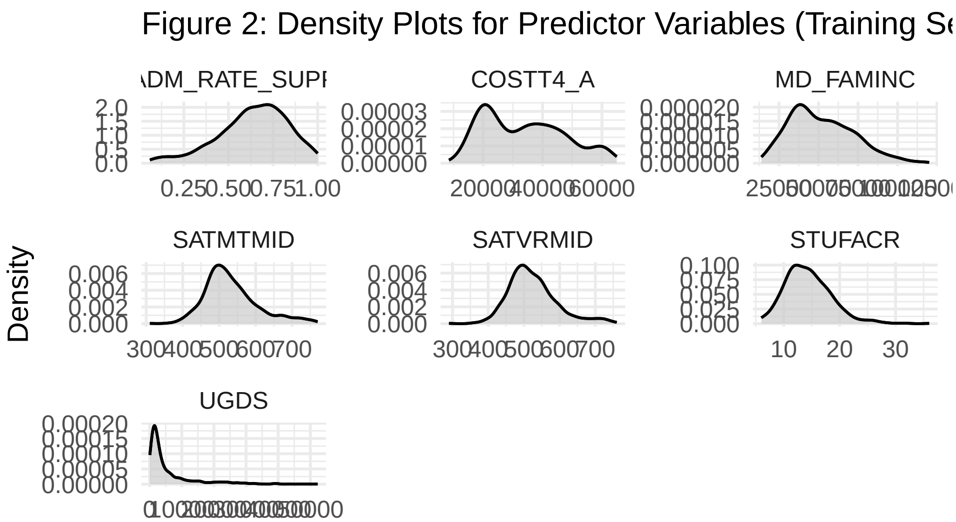
Variables:

Response Variable The response variable was median earnings ten years after enrollment (MD_EARN_WNE_P10), measured in U.S. dollars. This variable captures the median annual earnings of former students who received federal financial aid and are working ten years after initial enrollment.

Explanatory Variables The following institutional characteristics were examined as potential predictors of long-term earnings: Admissions rate (ADM_RATE): proportion of applicants admitted Median SAT Math score (SATMTMID) Median SAT Verbal score (SATVRMID) Total undergraduate enrollment (UGDS) Average cost of attendance (COSTT4_A), including tuition, fees, books, and room and board Student–faculty ratio (STUFACR) Median family income (MD_FAMINC) of students attending the institution

Statistical Methods: We analyzed the relationship between institutional characteristics and median earnings ten years after enrollment using multiple linear regression. All analyses were conducted on a randomly selected training subset of institutions with complete data. Initial exploratory analyses utilized scatterplots, correlation matrices, and diagnostic plots to assess distributional properties, linearity, and potential multicollinearity among predictors. Model development proceeded in stages. First, a full ("kitchen sink") model including all candidate predictors was estimated to identify potentially meaningful predictors and assess multicollinearity using variance inflation factors (VIFs). Due to high collinearity between median SAT Math and Verbal scores, only one SAT measure was retained in subsequent models. Model selection was guided by parsimony, Mallows' Cp, adjusted R^2, and nested F-tests, resulting in a final three-predictor model that included median math SAT, cost of attendance, and median family income. Model assumptions were evaluated for the final selected model using residual diagnostics. Potentially influential observations were assessed using Cook's distance, with outliers retained unless they exerted disproportionate influence on model estimates. To address potential nonlinearity in cost of attendance, bootstrapped coefficient estimates were used to assess the robustness of key predictors. Finally, model performance was evaluated using a held-out test set, and the fitted model was used to generate predicted earnings for selected institutions to illustrate practical application. Detailed exploratory analyses, diagnostics, and robustness checks are reported in subsequent sections.

```r
vars_to_plot <- c("MD_FAMINC", "STUFACR", "ADM_RATE_SUPP",
                  "SATVRMID", "SATMTMID", "COSTT4_A", "UGDS")

plot_long_filtered <- plot_long |>
  dplyr::filter(Variable %in% vars_to_plot)

# Figure 2: Density Plots for Predictor Variables
ggplot(plot_long_filtered, aes(x = Value)) +
  geom_density(fill = "grey80", alpha = 0.7, color = "black") +
  geom_density() +
  facet_wrap(~ Variable, scales = "free", ncol = 3) +
  labs(title = "Figure 2: Density Plots for Predictor Variables (Training Set)",
       x = "", y = "Density") +
  scale_y_continuous(labels = function(x) format(x, scientific = FALSE)) +
  theme_minimal() +
  theme(strip.text = element_text(size=9),
        plot.title = element_text(size=12))
```




```r
# Table 2: SAT Math and Verbal Statistics
favstats(SATMTMID ~ 1, data = college_train)
```

```
##   1 min  Q1 median    Q3 max     mean       sd   n missing
## 1 1 310 485    520 568.5 770 532.9477 73.77013 879       0
```

```r
favstats(SATVRMID ~ 1, data = college_train)
```

```
##   1 min  Q1 median    Q3 max     mean       sd   n missing
## 1 1 290 480    515 559.5 760 524.6576 68.93936 879       0
```

Results: Univariate EDA Response Variable Summary: Median Earnings 10 Years After Enrollment is unimodal and roughly normally distributed, with a rightward skew. The median is 43,600 and the mean is 45,821.73. The middle 50% of colleges have earnings between 38400 and 50100 (IQR approximately 13,000), with a large standard deviation of 12076.79, indicating substantial variations in Median 10 Year Earnings between colleges. There are several potential outliers on the right tail, which represent colleges specializing in high-earning professions, including pharmaceutical colleges and engineering universities.

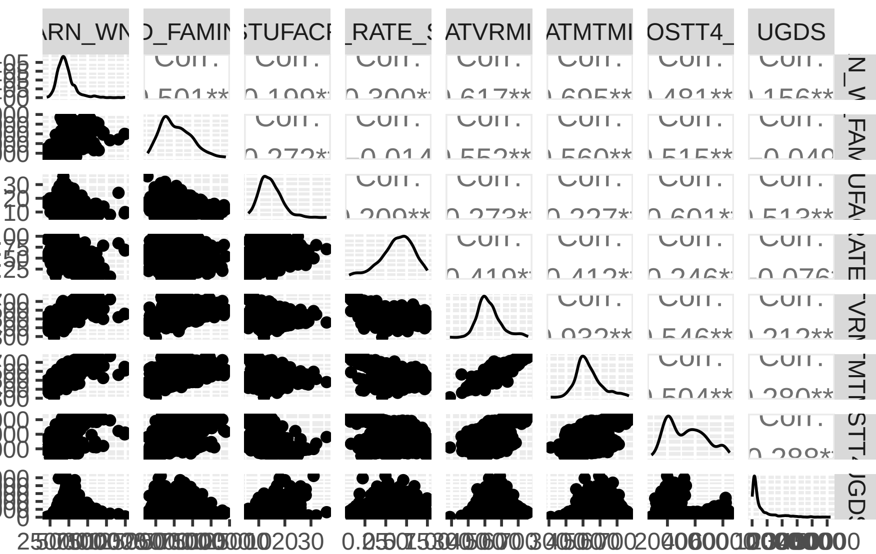
Right-Skewed Variables Several variables display a clear right skew, indicating a concentration of observations on the lower end and a long upper tail. This group includes Cost of Attendance, Median Family Income, Student–Faculty Ratio, Undergraduate Enrollment (certificate-seeking students). All predictor variables except average cost of attendance appear unimodal. Each shows potential high-end outliers, though formal assessment was not conducted at this stage. Average Cost of Attendance: trimodal; Median = 32,707; min = 8,461; max = 64,988; range = 56,527; IQR = 21,993.5. Median Family Income: Median = 49,259; min = 13,833; max = 120,000; range = 70,741; IQR = 29,391.5. Student–Faculty Ratio: Median = 14; range = 6 to 36 (range = 30); IQR = 6.

Left-Skewed Variable Only admissions rate shows left skew, with most schools admitting moderate to high percentages of applicants and fewer highly selective institutions. It is unimodal with no strong signs of outliers: Admission Rate: Median approx 0.66; Min = 0.0596 (approx6%), max = 1.00 (100%); Range = 0.94; IQR = 0.25

Approximately Normal Variables Two variables appear roughly symmetric, unimodal and bell-shaped, with no visible outliers: SAT Math: Median = 520; Mean = 532.95; range = 310–770; IQR = 83.5 SAT Verbal: Median = 515; Mean = 524.66; range = 290–760; IQR = 79.5

```r
library(GGally)
library(dplyr)

vars <- c("MD_EARN_WNE_P10", "MD_FAMINC", "STUFACR",
          "ADM_RATE_SUPP", "SATVRMID", "SATMTMID",
```

```
                "COSTT4_A", "UGDS")

train_pairs <- college_train %>%
  select(all_of(vars))

# Figure 3: GGpairs Plot of Response and Predictor Variables (Training Set)
ggpairs(train_pairs)
```



Bivariate EDA The graph above allowed us to examine the bivariate relationship between each explanatory variable and median earnings ten years after enrollment (MD_EARN_WNE_P10). We first produced a GGpairs plot to view scatterplots, density plots, and correlation coefficients among each of the 6 explanatory variables (ADM_RATE, SATVRMID, SATMTMID, UGDS, COSTT4_A, MD_FAMINC) and the response variable (MD_EARN_WNE_P10.) This provides an initial assessment of which institutional characteristics appear most strongly associated with post-graduation earnings, and it allows us to visually inspect distribution patterns and potential outliers. We also created graphs for each individual predictor and MD_EARN_WNE_P10.

Median Earnings 10 Years After Enrollment was positively correlated with Math and Verbal SAT scores (r = 0.695, r = 0.617, respectively). It was also positively correlated with median family income (r = 0.501), though we can see homoscedasticity. This may be partially due to the fact that most top academic schools have extensive scholarship opportunities for students of a range of incomes, and end up producing high earners. Additionally, MD_EARN_WNE_P10 was positively correlated with cost of attendance (r = 0.481). It was weakly negatively correlated with student-faculty ratio (r = -0.199) and admissions rate (r = -0.300) Though notably, for highly selective schools (admissions rate < 25%), there appears to be a moderate negative relationship where lower acceptance rates are correlated with higher future earnings. However, once the admissions rate exceeds roughly 30%, the slope becomes almost flat. This suggests that beyond a threshold of approximately 30%, institutional selectivity has a diminished association with median earnings ten years after enrollment.

The GGpairs matrix also helped us to identify multicollinearity among predictors, such as SAT verbal and SAT math scores (r = 0.932).

```
nrow(plot_data_complete)
```

```
## [1] 1173
```

```
nrow(college_train)
```

```
## [1] 879
```

```r
nrow(college_test)
```

```
## [1] 294
```

```r
y_train <- college_train$MD_EARN_WNE_P10
y_test  <- college_test$MD_EARN_WNE_P10

X_train <- college_train[, cols_to_plot[-1]]
X_test  <- college_test[, cols_to_plot[-1]]
```

```r
lmInitial <- lm(MD_EARN_WNE_P10 ~ ., data = college_train)

# Table 2: Initial Linear Model Summary
summary(lmInitial)
```

```
##
## Call:
## lm(formula = MD_EARN_WNE_P10 ~ ., data = college_train)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -23880  -4577   -943   3202  67887
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.921e+03  3.774e+03  -0.509  0.61081
## MD_FAMINC      8.993e-02  1.868e-02   4.813 1.75e-06 ***
## STUFACR        1.426e+02  9.551e+01   1.493  0.13592
## ADM_RATE_SUPP -4.339e+03  1.680e+03  -2.583  0.00995 **
## SATVRMID      -6.755e+01  1.158e+01  -5.833 7.65e-09 ***
## SATMTMID       1.353e+02  1.108e+01  12.211  < 2e-16 ***
## COSTT4_A       1.987e-01  3.130e-02   6.349 3.47e-10 ***
## UGDS           7.522e-02  4.940e-02   1.523  0.12820
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8234 on 871 degrees of freedom
## Multiple R-squared:  0.5388, Adjusted R-squared:  0.5351
## F-statistic: 145.4 on 7 and 871 DF,  p-value: < 2.2e-16
```

```r
# Table 4: Variance Inflation Factors for Initial Model
car::vif(lmInitial)
```

```
##     MD_FAMINC       STUFACR ADM_RATE_SUPP      SATVRMID      SATMTMID
##      1.852869      2.036980      1.383684      8.251630      8.650175
##      COSTT4_A          UGDS
##      2.347928      1.900669
```

Our Model: We began by running a "kitchen sink" model to determine the potentially significant predictors. From this, we saw that median family income (MD_FAMINC), admissions rate (ADM_RATE_SUPP) SAT verbal and math (SATVRMID, SATMTMID) and cost of attendance (COSTT4_A) are all potentially useful in predicting median income 10 years after enrollment. Neither undergraduate size (UGDS) nor student-faculty ratio (STUFACR) were significant.

Next, we ran a VIF in order to determine if any multicollinearity is present. The only major concern was between SATVRMID and SATMTMID, where their correlation was 0.932 and both had VIFs above 8.

In order to determine the best predictors to use for our model, we ran a stepwise regression. To determine the best model, we considered parsimony, Mallows' Cp, the adjusted R^2 and the overall F-statistic. Our chosen model had a relatively low Mallow's CP, few predictors, and a relatively high adjusted R-squared of 0.521210. However, this model included SATVRMID, SATMTMID, COSTT4_A, MD_FAMINC . Since we saw previously that SATVRMID and SATMTMID were multicollinear predictors (VIF>5), we chose to keep only SATTMID (Median Math SAT), as this was a stronger predictor of future earnings than the Median Verbal SAT score.

Then, we ran a nested-F test, with four separate models. We determined that the strongest model included SATMTMID, COSTT4_A, and MD_FAMINC. With an adjusted R-squared value of 0.513, our model accounts of 51.3% of variability in Median Earnings 10 Years After Enrollment (F-statistic: 309 on 3 and 875 DF, p-value: <2e-16)

```
require(leaps)
```

```
## Loading required package: leaps
```

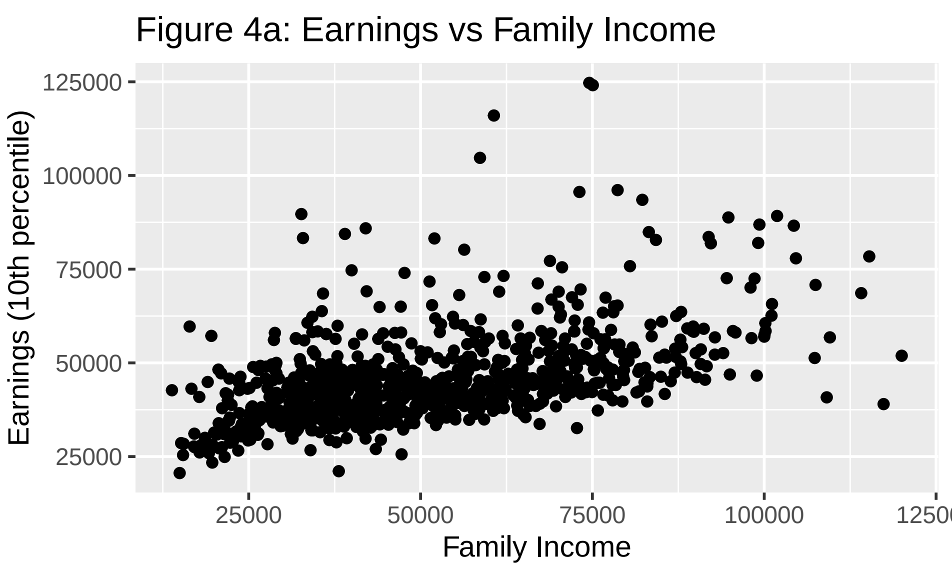
```
stepwise <- regsubsets(MD_EARN_WNE_P10 ~ .,
                       data = college_train,
                       method = "seqrep",
                       nbest = 1)

# Table 5: Stepwise Regression Model Selection Metrics
with(summary(stepwise), data.frame(rsq, adjr2, cp, rss, outmat))
```

```
##                 rsq     adjr2         cp         rss MD_FAMINC STUFACR
## 1  ( 1 ) 0.4833875 0.4827984 100.669975 66154965409
## 2  ( 1 ) 0.5063467 0.5052196  59.309521 63214924833
## 3  ( 1 ) 0.5228463 0.5212104  30.148380 61102054410
## 4  ( 1 ) 0.4286192 0.4260042 210.104958 73168338703         *       *
## 5  ( 1 ) 0.5344884 0.5318223  12.161173 59611219486         *
## 6  ( 1 ) 0.5376302 0.5344488   8.227688 59208901534         *
## 7  ( 1 ) 0.5388097 0.5351033   8.000000 59057853950         *       *
##          ADM_RATE_SUPP SATVRMID SATMTMID COSTT4_A UGDS
## 1  ( 1 )                              *
## 2  ( 1 )                              *        *
## 3  ( 1 )                     *        *        *
## 4  ( 1 )            *        *
## 5  ( 1 )                     *        *        *    *
## 6  ( 1 )            *        *        *        *    *
## 7  ( 1 )            *        *        *        *    *
```

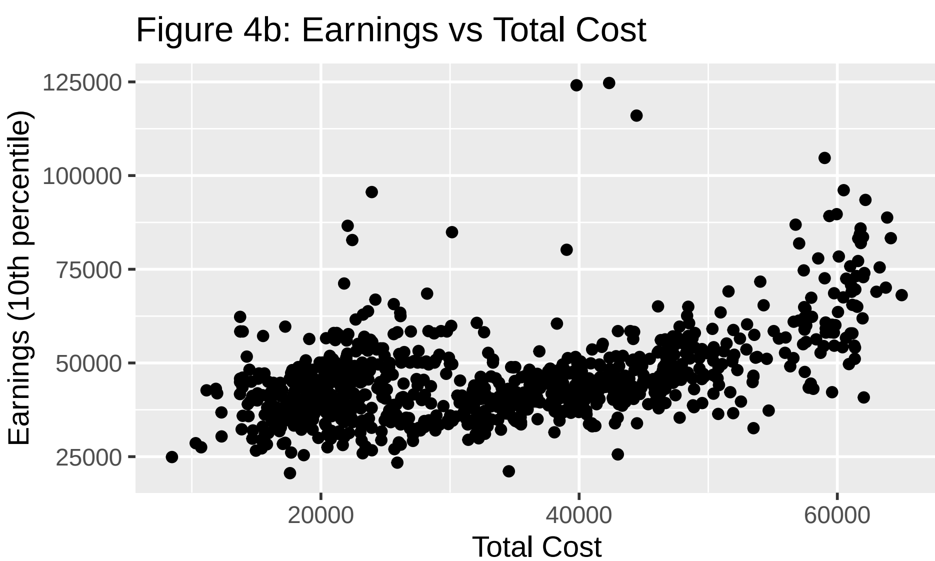
```
lmPost <- lm(MD_EARN_WNE_P10 ~ SATMTMID + COSTT4_A, data = college_train)

# Table 6: VIFs for Reduced Model
car::vif(lmPost)
```

```
## SATMTMID COSTT4_A
## 1.340246 1.340246
```

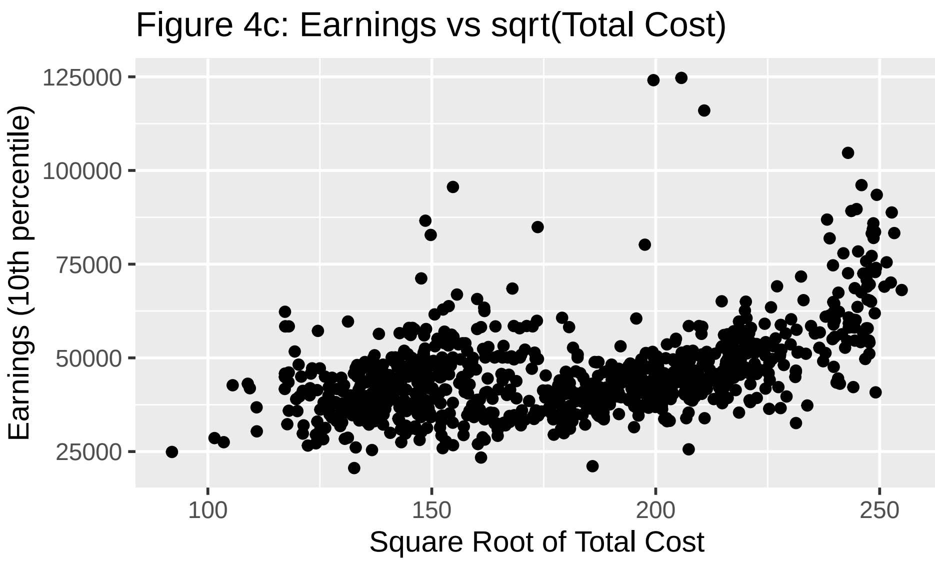
```
# Figures 4a-f: Earnings vs Predictors
gf_point(MD_EARN_WNE_P10 ~ MD_FAMINC, data = college_train) %>%
  gf_labs(title="Figure 4a: Earnings vs Family Income", x="Family Income",
          y="Earnings (10th percentile)")
```



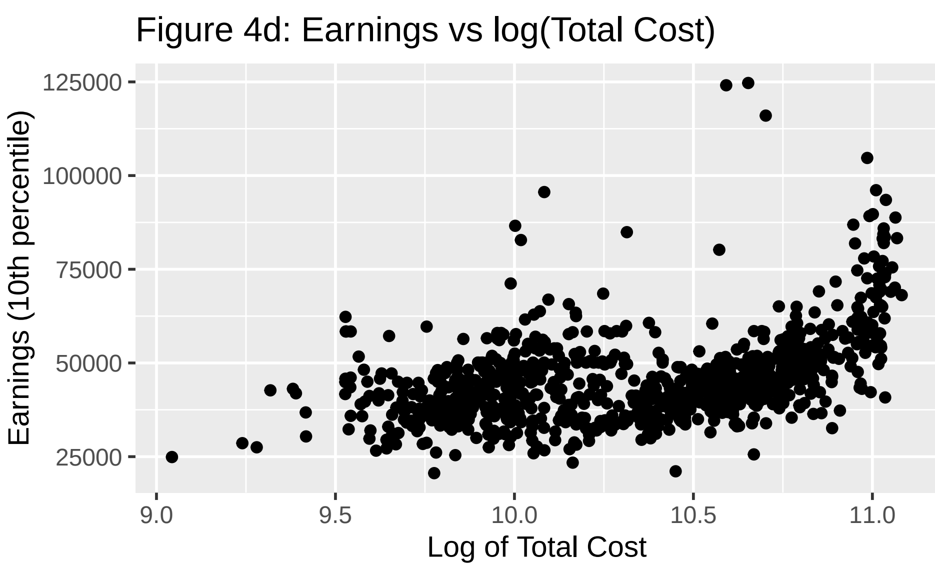

```r
gf_point(MD_EARN_WNE_P10 ~ COSTT4_A, data = college_train) %>%
  gf_labs(title="Figure 4b: Earnings vs Total Cost", x="Total Cost",
          y="Earnings (10th percentile)")
```




```r
gf_point(MD_EARN_WNE_P10 ~ sqrt(COSTT4_A), data = college_train) %>%
  gf_labs(title="Figure 4c: Earnings vs sqrt(Total Cost)", x="Square Root of Total Cost",
          y="Earnings (10th percentile)")
```

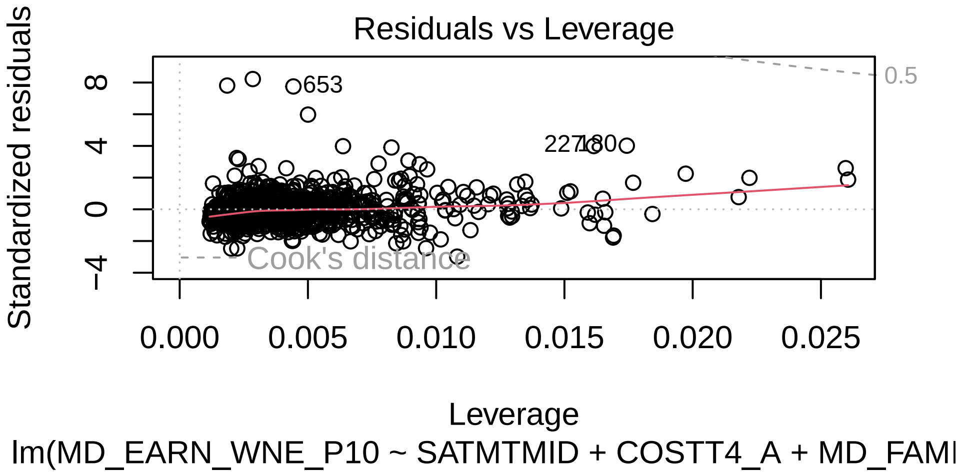



```r
gf_point(MD_EARN_WNE_P10 ~ log(COSTT4_A), data = college_train) %>%
  gf_labs(title="Figure 4d: Earnings vs log(Total Cost)", x="Log of Total Cost",
          y="Earnings (10th percentile)")
```



Unusual Points: Given that our upper tail on our QQ plot diverged strongly from the line, and that our data for several variables was not normally distributed, we decided to investigate unusual points. We first looked at the plot that showed Cook's Distance, and Residuals vs. Leverage. From this combined plot (shown above), we concluded that no points exceed a Cook's distance of 0.5, indicating that no single institution exerted undue influence on the fitted regression model. Notably, though, after calculating the high-leverage cut off value of 0.0113766, we found that there were 63 high leverage points. However, we decided not to eliminate these from the model due to the fact that none exceeded Cook's Distance values of 0.5. Notably, the three numbered points on the QQ plot were all pharmaceutical colleges, with high acceptance rates, average SAT scores, and very high median earnings (all above $100,000). After looking into this phenomenon, we discovered that in recent decades, it has become relatively easier to get into many U.S. pharmacy colleges due to a significant increase in the number of programs and a decline in applicants. Notably, the profession still generally offers a high starting salary for graduates, in part due to a sustained national reliance on pharmaceuticals.

```r
# Figure5) Optional diagnostic plots

fm3 <- lm(MD_EARN_WNE_P10 ~ SATMTMID + COSTT4_A + MD_FAMINC, data = college_train)

plot(fm3, which = 5)
```




```r
lev <- hatvalues(fm3)

# Table 7) Leverage diagnostics (training data only)
# Cook's Distance & High-Leverage Points
p <- length(coef(fm3)) - 1
n <- nrow(model.frame(fm3))



lev_cutoff <- 2 * (p + 1) / n
lev_cutoff
```

```
## [1] 0.009101251
```

```r
sum(lev > lev_cutoff)
```

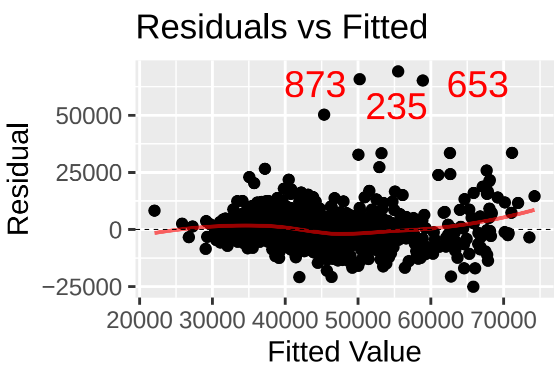
```
## [1] 69
```

Conditions

In our bivariate exploratory data analysis, we found that SAT scores were strongly linearly correlated with median earnings, and met all conditions, so we did not attempt any transformation with this variable. However, we attempted several transformations of median family income (MD_FAMINC) and cost of attendance (COSTT4_A) to ameliorate the upper tails of our final model's QQ plot above. In our bivariate EDA, both variables exhibited homogeneity and cost of attendance appeared almost parabolic. Unfortunately, square root and log transformations were unsuccessful in improving conditions. Checking our conditions, we see that our Residuals vs Fitted plot appears to be roughly centered about 0, with no visible heteroskedasticity. Thus, we can conclude that linearity, homoschedasticity, and independence of errors conditions are met. Looking at our QQ-plot, we see that the lower and middle parts of the data are relatively normal, however our upper tail contains several influential points that skew the data upwards off of the line. This tells us that the model has unusually large residuals, and tells us that they are not normally distributed. We attempted to address this issue with transformations and an analysis of unusual points, though we were largely unsuccessful at linearizing our data.

```r
# Figure 6: Residuals vs Fitted for Final Model
lmClose <- lm(MD_EARN_WNE_P10 ~ SATMTMID + COSTT4_A + MD_FAMINC, data = college_train)
mplot(lmClose, which=1)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



```r
# Table 8: Nested Model Comparison
fm1 <- lm(MD_EARN_WNE_P10 ~ SATMTMID, data = college_train)
fm2 <- lm(MD_EARN_WNE_P10 ~ SATMTMID + COSTT4_A, data = college_train)
fm3 <- lm(MD_EARN_WNE_P10 ~ SATMTMID + COSTT4_A + MD_FAMINC, data = college_train)
fm4 <- lm(MD_EARN_WNE_P10 ~ SATMTMID + COSTT4_A + MD_FAMINC + ADM_RATE_SUPP, data = college_train)
anova(fm1, fm2, fm3, fm4)
```

```
## Analysis of Variance Table
##
## Model 1: MD_EARN_WNE_P10 ~ SATMTMID
## Model 2: MD_EARN_WNE_P10 ~ SATMTMID + COSTT4_A
## Model 3: MD_EARN_WNE_P10 ~ SATMTMID + COSTT4_A + MD_FAMINC
## Model 4: MD_EARN_WNE_P10 ~ SATMTMID + COSTT4_A + MD_FAMINC + ADM_RATE_SUPP
##   Res.Df        RSS Df  Sum of Sq       F    Pr(>F)
## 1    877 6.6155e+10
## 2    876 6.3215e+10  1 2940040576 41.4597 1.983e-10 ***
## 3    875 6.2163e+10  1 1052165481 14.8374 0.0001258 ***
## 4    874 6.1978e+10  1  184554003  2.6025 0.1070542
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

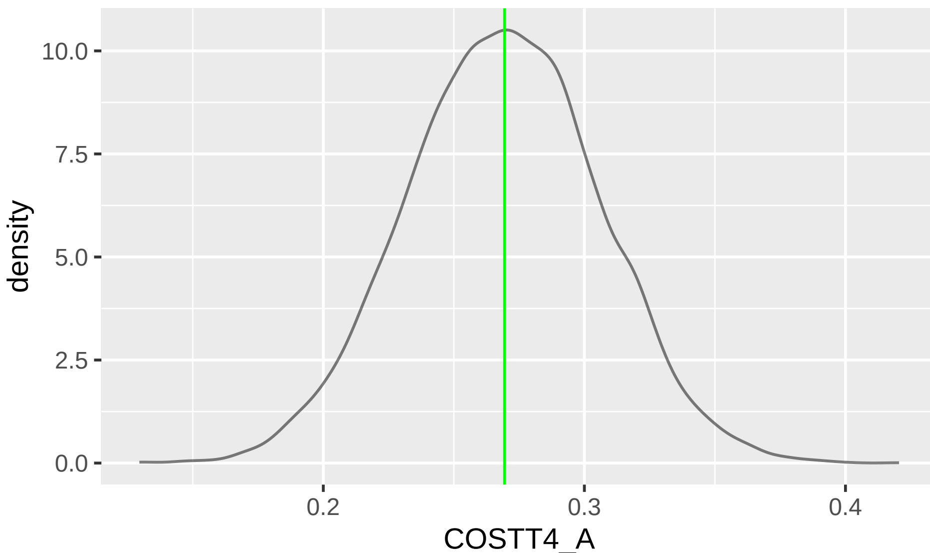
```r
# Table 9: Bootstrap Coefficients (Single Draw)
sim1 <- college_train[sample(nrow(college_train), replace=TRUE), ]
m.tmp1 <- lm(MD_EARN_WNE_P10 ~ COSTT4_A + MD_FAMINC, data=sim1)
coef(m.tmp1)["COSTT4_A"]
```

```
##  COSTT4_A
## 0.2170839
```

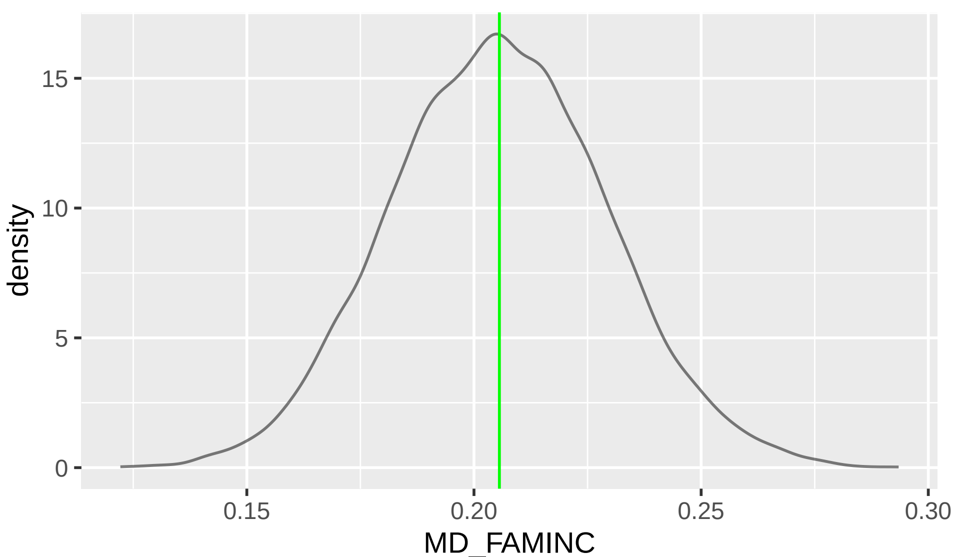
```r
coef(m.tmp1)["MD_FAMINC"]
```

```
## MD_FAMINC
## 0.2375404
```

```r
# Figures 6-7: Bootstrap Distributions
set.seed(230)
actual_model <- lm(MD_EARN_WNE_P10 ~ COSTT4_A + MD_FAMINC, data = college_train)
coef.actual <- coef(actual_model)
B <- 10000
boot.coef <- replicate(B, {
  sim <- college_train[sample(nrow(college_train), replace=TRUE), ]
  coef(lm(MD_EARN_WNE_P10 ~ COSTT4_A + MD_FAMINC, data=sim))
})
boot.coef <- as.data.frame(t(boot.coef))

gf_dens(~ COSTT4_A, data=boot.coef) %>%
  gf_vline(xintercept = coef.actual["COSTT4_A"], color="green") # Figure 6
```




```r
gf_dens(~ MD_FAMINC, data=boot.coef) %>%
  gf_vline(xintercept = coef.actual["MD_FAMINC"], color="green") # Figure 7
```



```r
# Tables 9-10: 95% Bootstrap CI
quantile(boot.coef$COSTT4_A, c(0.025, 0.975))
```

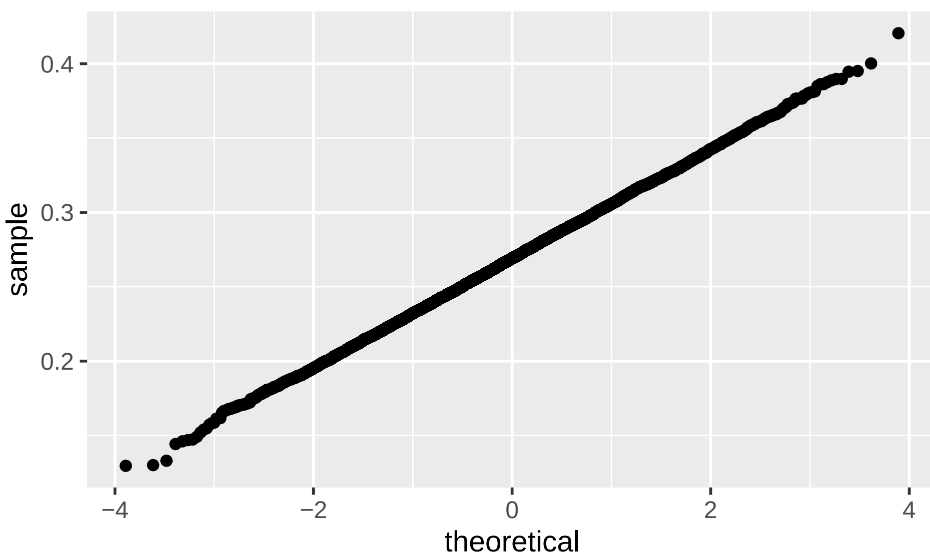
```
##      2.5%     97.5%
## 0.1967022 0.3403306
```

```r
quantile(boot.coef$MD_FAMINC, c(0.025, 0.975))
```

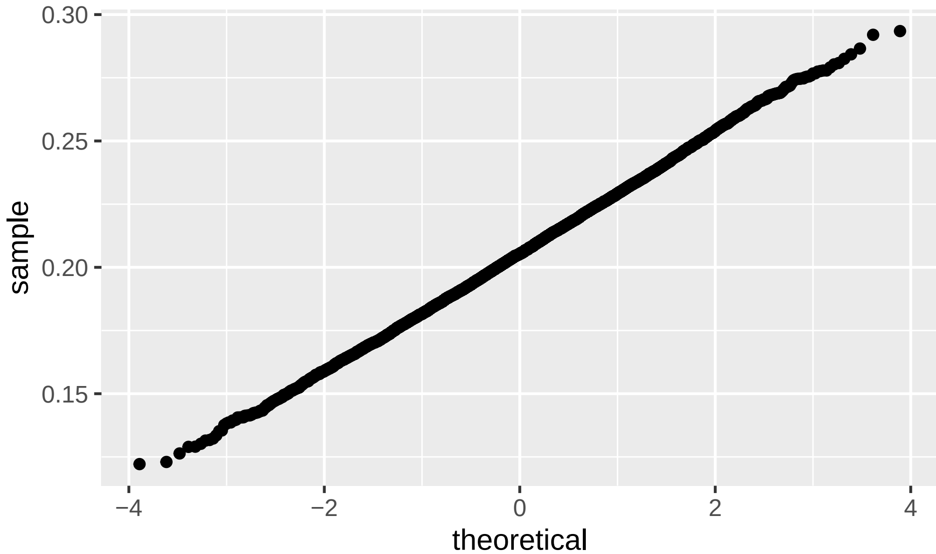
```
##      2.5%     97.5%
## 0.1598376 0.2529818
```

Bootstrapping Even though we applied transformations to the predictors and response and checked for unusual points,these adjustments did not fully correct issues such as skewness, non-linearity, and heteroscedasticity. As a result, standard regression assumptions were not entirely met, making traditional confidence intervals and p-values less reliable. To address this, we used bootstrapping, a resampling method that does not rely on these assumptions. By repeatedly resampling the training data and refitting the regression model 10,000 times, we generated empirical distributions of the coefficients for COSTT4_A and MD_FAMINC. These bootstrap distributions (Figures 6-7) and the corresponding 95% confidence intervals (Tables 9-10) allow us to quantify the uncertainty of our estimates in a robust, data-driven way. For median family income, we can be 95% confident that the true coefficient falls between 0.1598 and 0.25298. For cost of attendance, we can be 95% confident that the true coefficient falls between 0.1967022 and 0.34033. Overall, bootstrapping strengthens the credibility of our conclusions by providing reliable measures of variability and highlighting which predictors are meaningfully associated with post-graduation earnings.

```r
# Figures 10-11: QQ Plots for Bootstrap
gf_qq(~ boot.coef$COSTT4_A) # Figure 8
```




```r
gf_qq(~ boot.coef$MD_FAMINC) # Figure 9
```



Predictions:

Pred. Median Earnings 10 Years after Enrollment = -10,800+91.7(SATMTMID)+0.124(COSTT4_A)+0.069(MD_FAMINC) =

Using the final regression model trained on 2014 College Scorecard data, predicted median earnings ten years after enrollment were calculated for Amherst College and UMASS Amherst. For Amherst College, the model predicted median earnings of $66,565.20, based on a median Math SAT score of 725, an average cost of attendance of $61,544, and a median family income of $47,119.50. The observed median earnings value for Amherst graduates was $65,000, yielding a residual of -$1,565.20 (observed minus predicted), indicating that the model slightly overestimated earnings for Amherst students. For the University of Massachusetts Amherst, the model predicted median earnings of $54,412.43, using a median Math SAT score of 620, an average cost of attendance of $26,207, and a median family income of $74,040. The observed median earnings ten years post-enrollment for UMass graduates was $51,400, resulting in a residual of -$3,012.43, again reflecting a slight overprediction. In both cases the predicted values were reasonably close to the observed outcomes.

```
# Table 12: Training Model Summary and Performance
trainfm <- lm(MD_EARN_WNE_P10 ~ SATMTMID + COSTT4_A + MD_FAMINC, data=college_train)
msummary(trainfm)
```

```
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.082e+04  2.148e+03  -5.039 5.68e-07 ***
## SATMTMID     9.173e+01  4.883e+00  18.786  < 2e-16 ***
## COSTT4_A     1.236e-01  2.561e-02   4.826 1.64e-06 ***
## MD_FAMINC    6.902e-02  1.794e-02   3.848 0.000128 ***
##
## Residual standard error: 8429 on 875 degrees of freedom
## Multiple R-squared:  0.5146, Adjusted R-squared:  0.5129
## F-statistic: 309.2 on 3 and 875 DF,  p-value: < 2.2e-16
```

```
train_r2 <- rsquared(trainfm)
predvalues <- predict(trainfm, newdata=college_test)
cvcor <- cor(predvalues ~ MD_EARN_WNE_P10, data=college_test)
train_r2
```

```
## [1] 0.5145631
```

```
cvcor^2
```

```
## [1] 0.5486228
```

```
train_r2 - cvcor^2
```

```
## [1] -0.0340596
```

Cross-Validation: To assess out-of-sample performance, we randomly split the complete-case dataset into a training set (879 institutions; 75%) and a test set (294 institutions; 25%). The model predicting median earnings ten years after enrollment (MD_EARN_WNE_P10) from SAT Math scores (SATMTMID), cost of attendance (COSTT4_A), and median family income (MD_FAMINC was fit using the training data, yielding a training R^2=0.515. We then generated predictions for the test set and computed the correlation between predicted and observed earnings, which was 0.741. Squaring this correlation produced a cross-validated performance estimate of 0.549. Shrinkage – the difference between the training R2 and the cross-validated R2, was -0.034, indicating no evidence of overfitting. This suggests that the model generalizes well across subsets of the data. Although negative shrinkage at first appeared counterintuitive, it can arise due to random sampling variability when the test set exhibits slightly less noise or a stronger linear relationship than the training set. However, this result should be interpreted cautiously because cross-validation was performed on a complete-case sample; if missingness in key variables is not completely random, the validation sample may overrepresent more well-documented institutions, potentially inflating apparent predictive performance and understating true generalization error in the broader population.

Conclusions:

This study investigated which institutional characteristics of U.S. colleges and universities best predict graduates' median earnings ten years after enrollment using College Scorecard data. We found that median SAT Math scores, average cost of attendance, and median family income were significant predictors of long-term earnings, collectively explaining approximately 51.5% of the variation in median earnings. A 75/25 holdout cross-validation indicated that the model generalized well to new data, with little evidence of overfitting. Diagnostic analyses revealed no strongly influential observations, which suggests that the estimated relationships are stable within the analyzed sample.

There are several limitations within these findings. The analysis was restricted to a complete-case sample, excluding institutions with missing data. If missingness is systematic, the results may overstate predictive performance and limit generalizability. In addition, all predictors were measured at the institutional level, and the model does not account for individual student characteristics, fields of study, regional labor markets, or graduate education. All of these factors influence earnings outcomes. As a result, the observed relationships should not be interpreted as causal effects or as predictions for individual students. Finally, although the model explains a substantial portion of earnings variation, nearly half of the variability remains unexplained.

References

Coughter, J., & Gowder, C. (2024, September 19). Why is the cost of college rising so fast? SSTI. https://ssti.org/blog/why-cost-college-rising-so-fast SSTI

Hanson, M. (2025, August 8). Student loan debt statistics. EducationData.org. https://educationdata.org/student-loan-debt-statistics Education Data Initiative

U.S. Department of Education. (n.d.). CollegeScorecard. https://www.ed.gov